
1.23
Ring
Enum

Implementing a VI

TOPICS

A. Designing Controls and Indicators
B. LabVIEW Data Types
C. While Loop
D. For Loop
E. Timing a VI
F. Data Feedback in Loops
G. Case Structure

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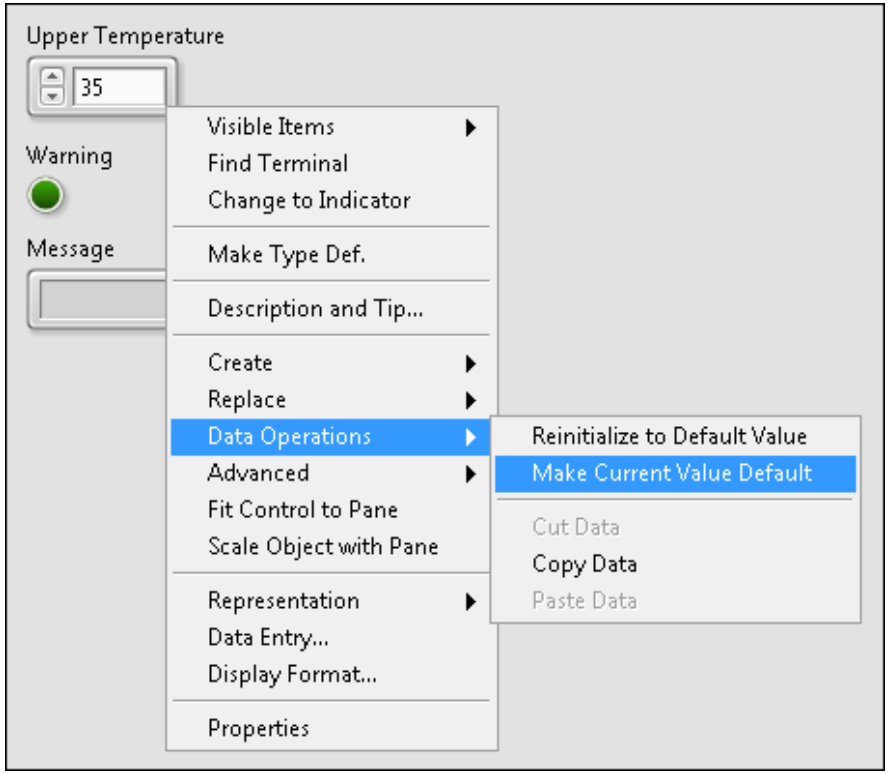
A. Designing Controls and Indicators

Labels & Options

Make sure to label controls and indicators clearly and set a default value if necessary.







B. LabVIEW Data Types

Shortcut Menu and Properties Dialog Box

Numeric Types

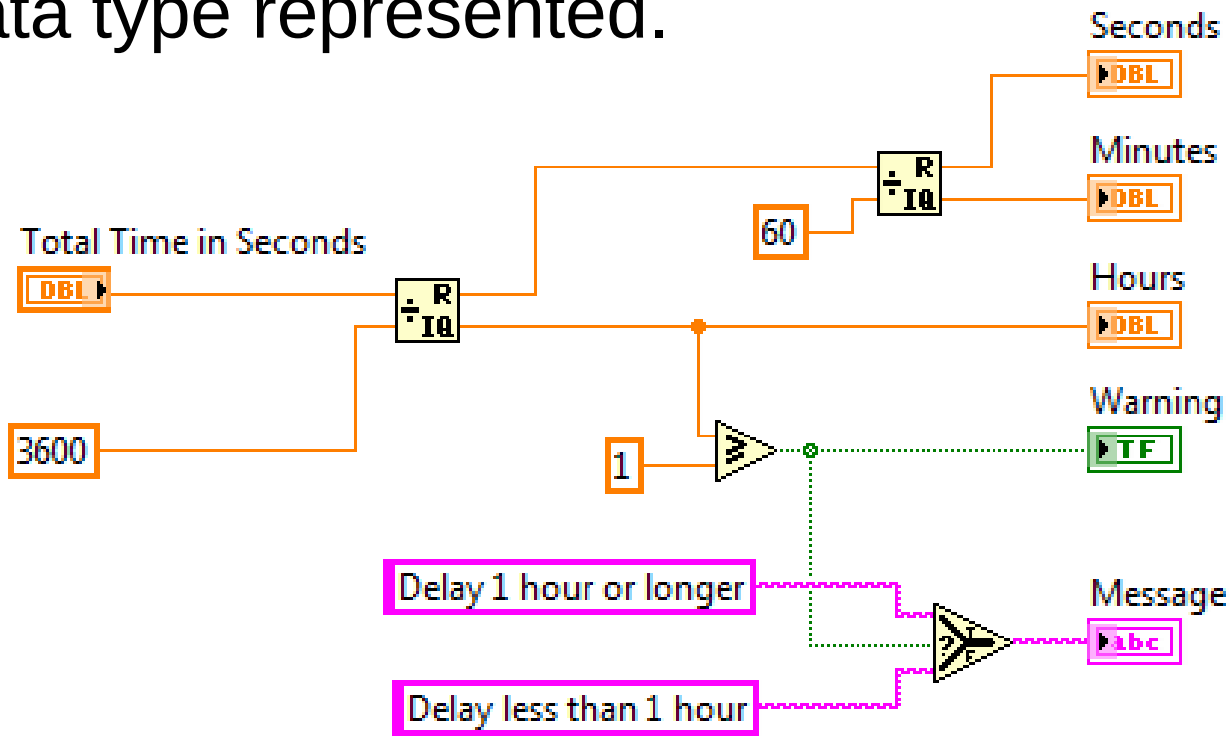
Boolean Types

String Types

Enums and Other Types

LabVIEW Data Types

Terminals visually communicate information about the data type represented.



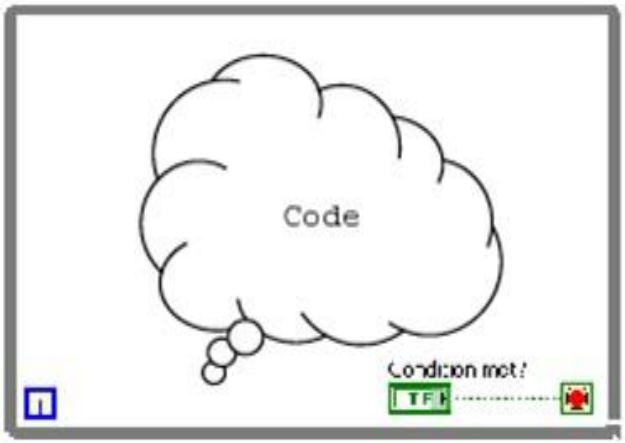
C. While Loop

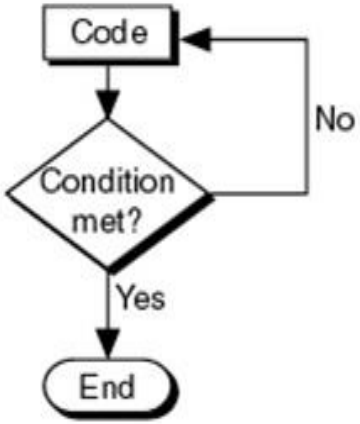
Iteration and Conditional Terminals

Tunnels

Error Checking

While Loop





Repeat (code);

Until Condition met;

End;

LabVIEW While Loop

Flowchart

Pseudo Code

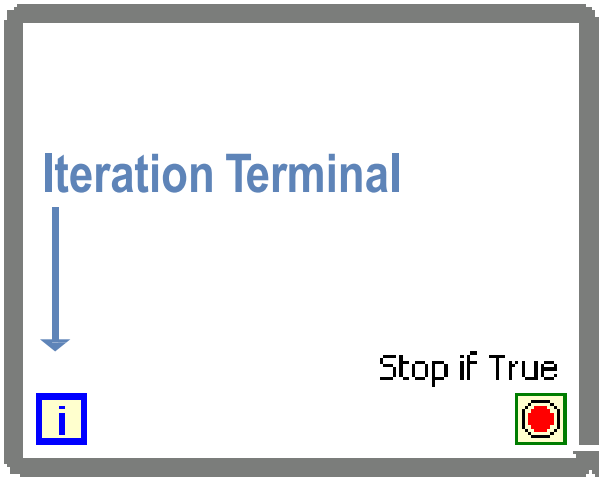
While Loop

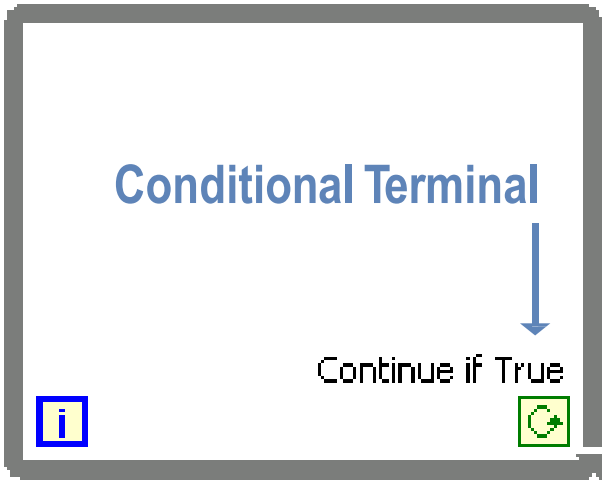
Iteration terminal

- Returns number of times loop has executed.
- Is zero-indexed.

Conditional terminal

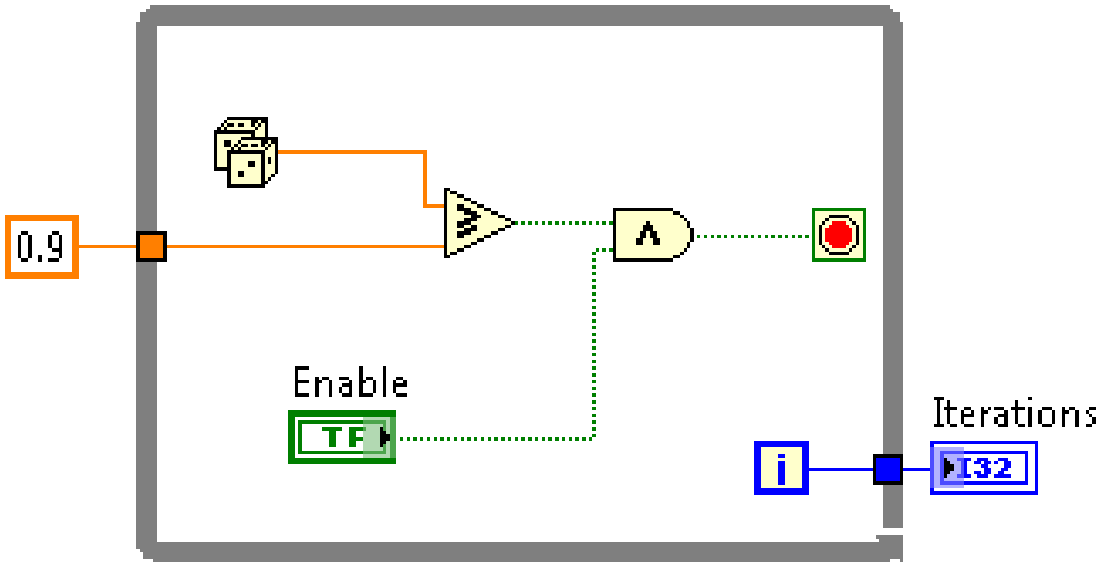
- Defines when the loop stops.
- Has two options:
  - Stop if True
  - Continue if True.





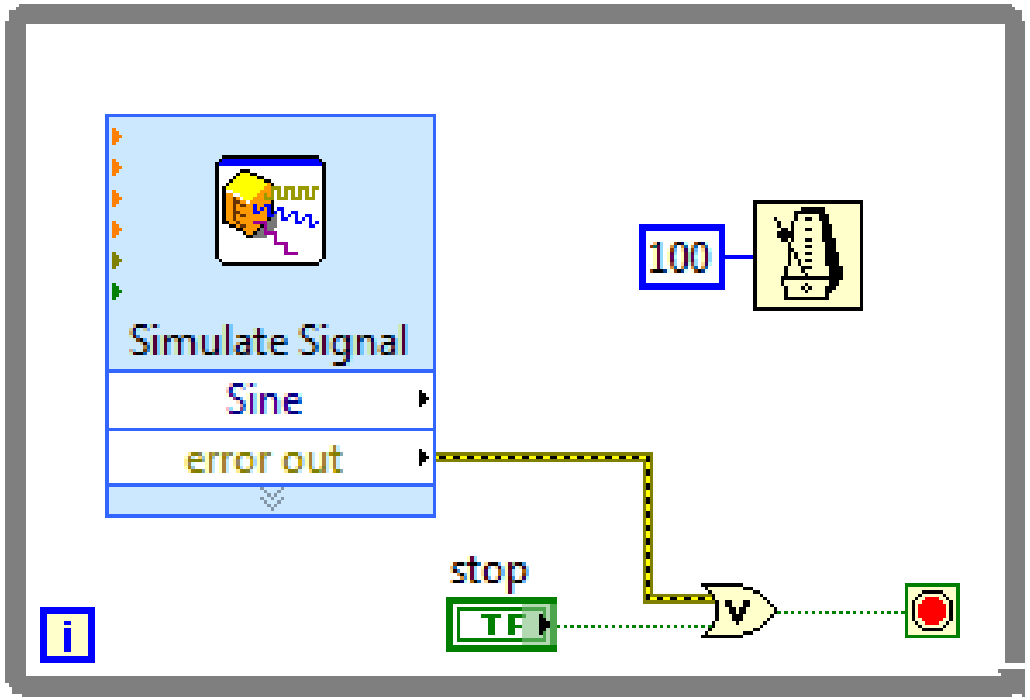
While Loop – Tunnels

- Tunnels transfer data into and out of structures.
- When a tunnel passes data into a loop, the loop executes only after data arrive at the tunnel (at all tunnels, if there is more than one).
- Data pass out of a loop after the loop terminates.



While Loop – Error Checking and Error Handling

Use an error cluster in a While Loop to stop the While Loop if an error occurs.



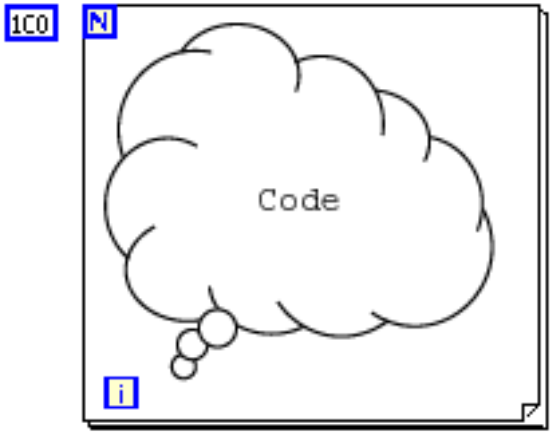
D. For Loop

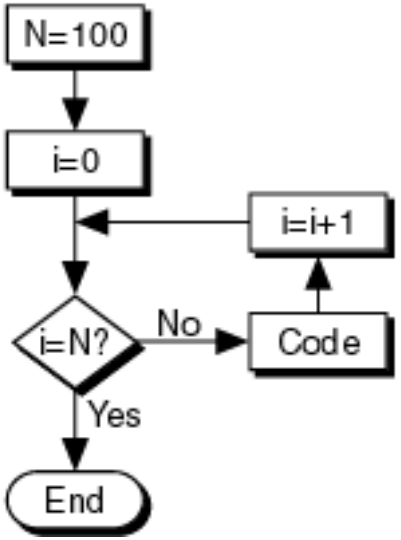
Conditional Terminal

Comparison with While Loop

Numeric Conversion for Count Terminal

For Loop



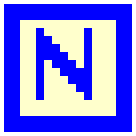



```
N=100;
i=0;
Until i=N:
    Repeat (code;i=i+1);
End;
```

LabVIEW For Loop | Flowchart | Pseudo Code

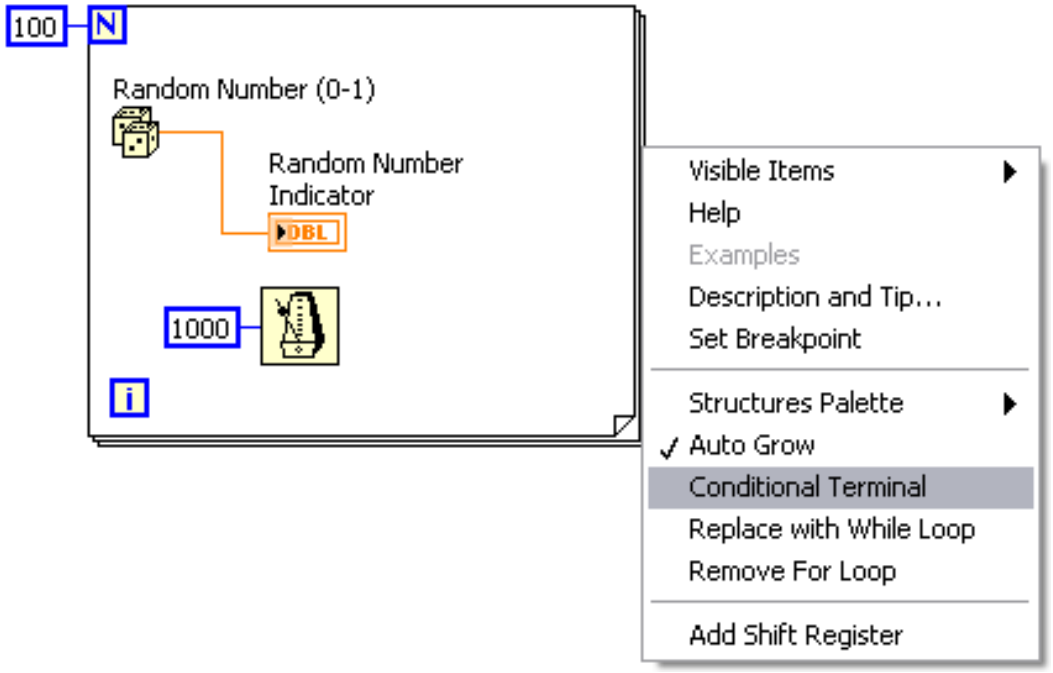
For Loop

- Create a For Loop the same way you create a While Loop.
- You can replace a While Loop with a For Loop by right-clicking the border of the While Loop and selecting **Replace with For Loop** from the shortcut menu.
- The value in the count terminal (an input terminal) indicates how many times to repeat the subdiagram in the For Loop.



For Loop – Conditional Terminal

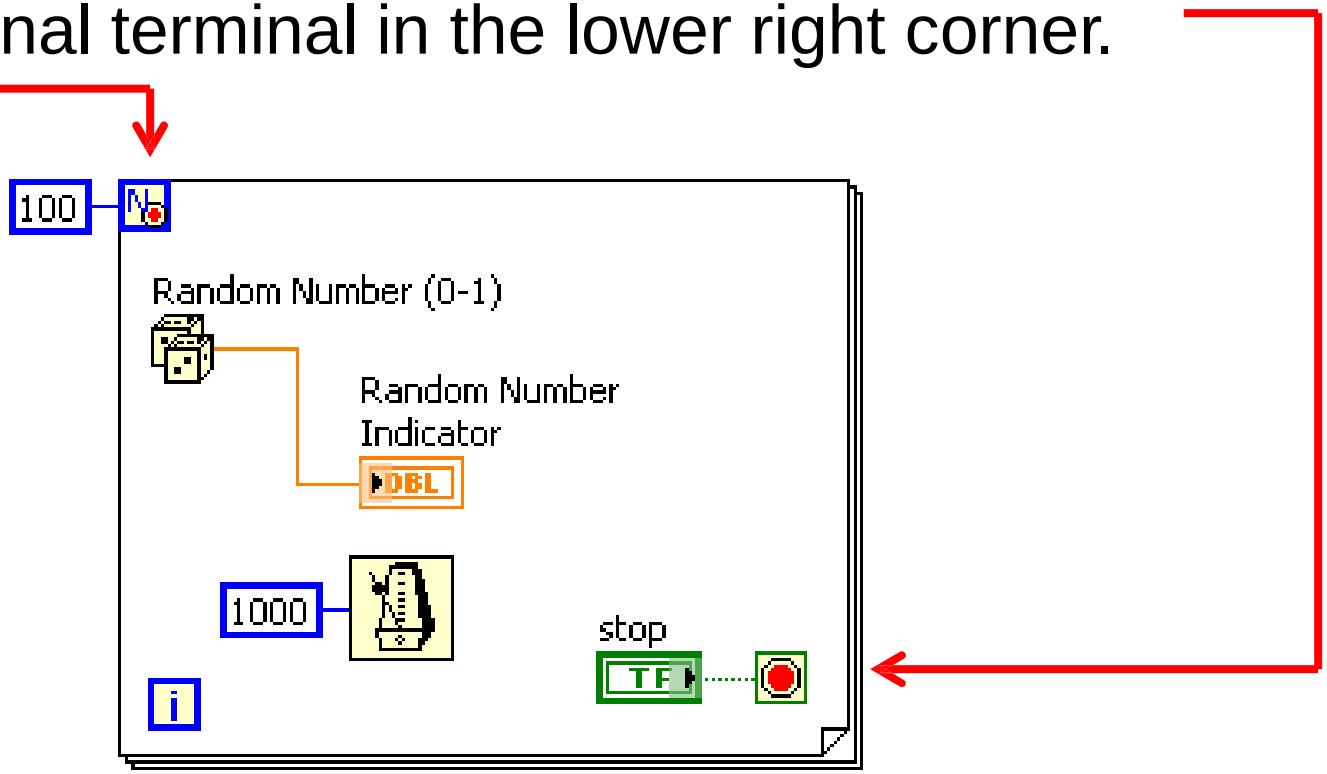
You can add a conditional terminal to configure a For Loop to stop when a Boolean condition is true or an error occurs. This is similar to a ‘break’ statement in text-based programming languages.



For Loop – Conditional Terminal

For Loops configured with a conditional terminal have:

- A red glyph next to the count terminal
- A conditional terminal in the lower right corner.



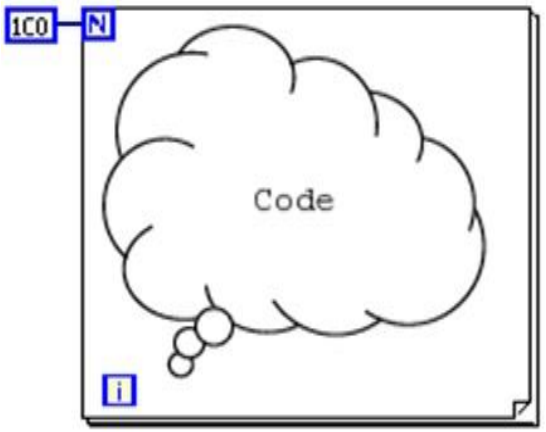
For Loop / While Loop Comparison

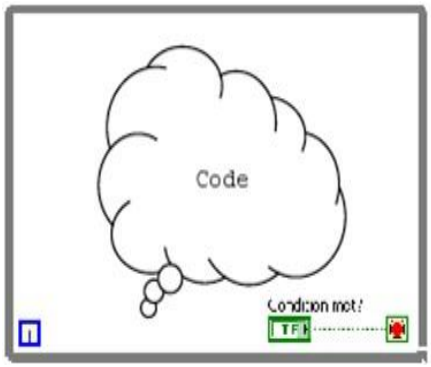
For Loop

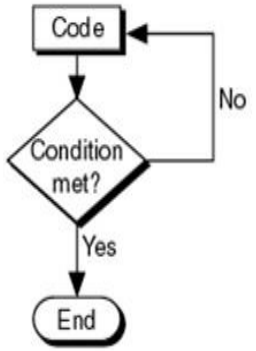
- Executes a set number of times unless a conditional terminal is added.
- Can execute zero times.
- Tunnels automatically output an array of data.

While Loop

- Stops executing only if the value at the conditional terminal meets the condition.
- Must execute at least once.
- Tunnels automatically output the last value.



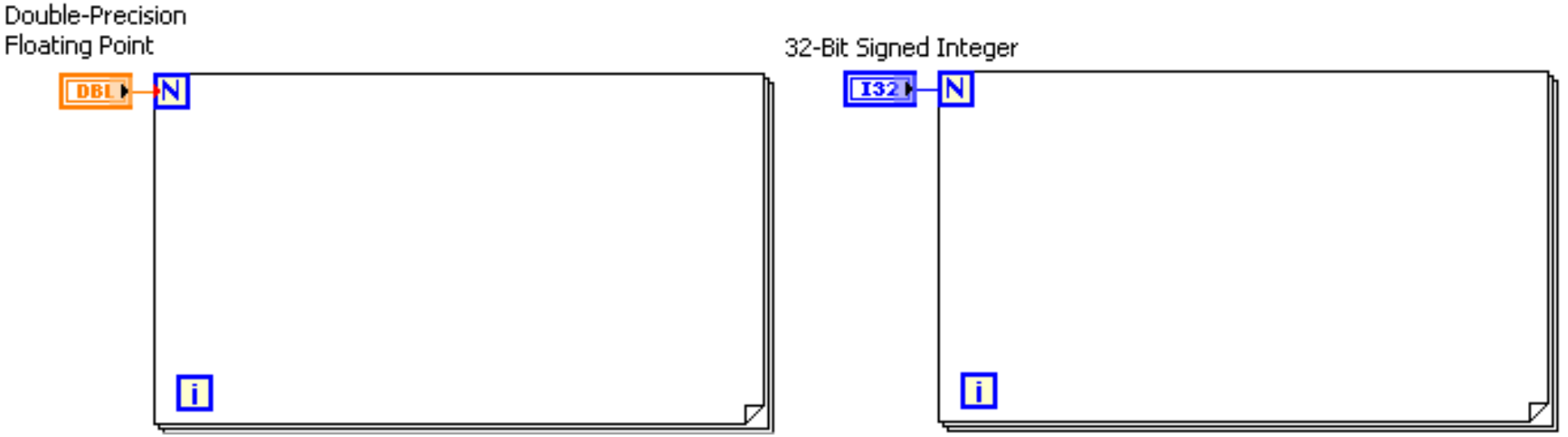




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For Loop – Numeric Conversion

- The number of iterations a For Loop executes must be specified in non-negative integers.
- If you wire a double-precision, floating-point numeric value to the count terminal, LabVIEW converts the numeric value to a 32-bit signed integer.



E. Timing a VI

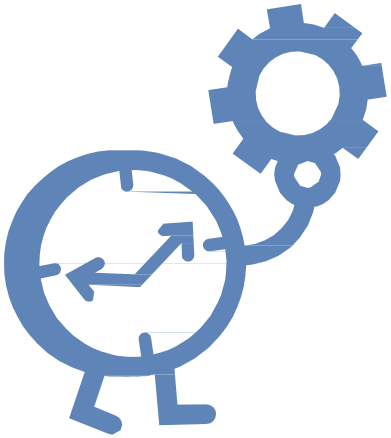
Reasons To Use Timing

Wait Functions and Express VIs

Timing a VI

Why do you need timing in a VI?

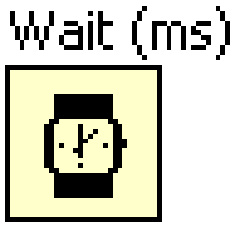
- To control the frequency at which a loop executes.
- To provide the processor with time to complete other tasks, such as processing the user interface or running other applications.



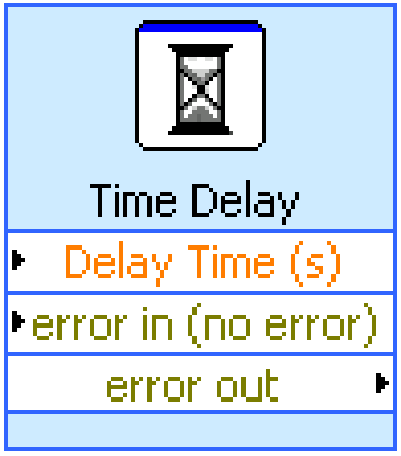
Wait Functions

A wait function inside a loop:

- Allows the VI to sleep for a set amount of time.
- Allows the processor to address other tasks during the wait time.
- Uses the operating system millisecond clock.







Elapsed Time Express VI

- Determines how much time elapses after some point in your VI.
- Keeps track of time while the VI continues to execute.
- Does not provide the processor with time to complete other tasks.



F. Data Feedback in Loops

Shift Registers

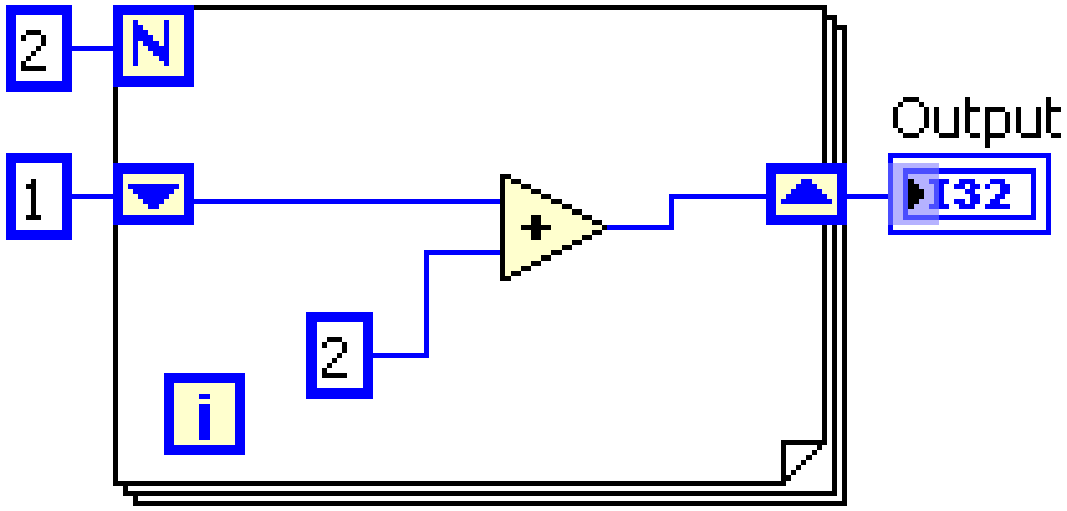
Initializing Shift Registers

Default for Unwired Values

Compound Shift Registers

Data Feedback in Loops

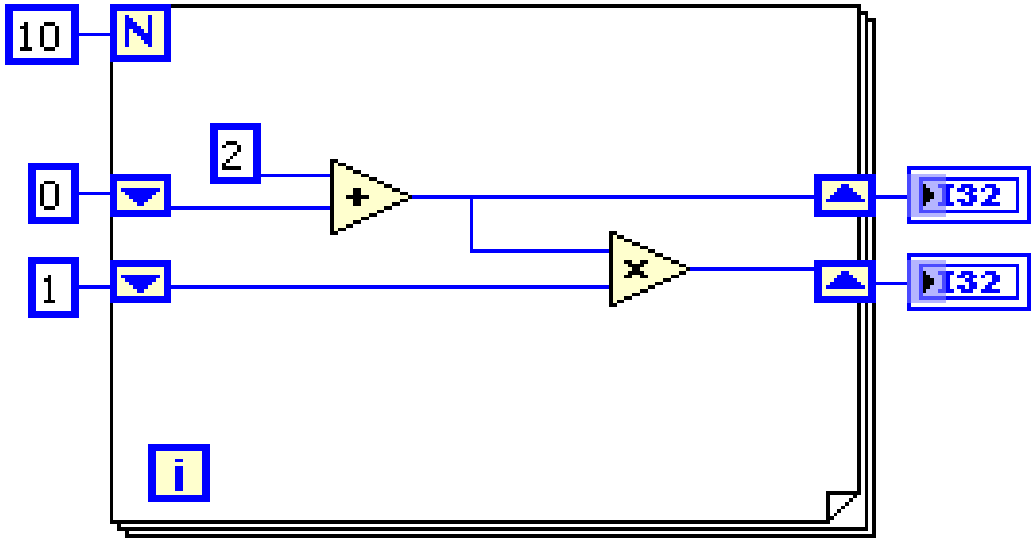
- When programming with loops, you often need to know the values of data from previous iterations of the loop.
- Shift registers transfer values from one loop iteration to the next.



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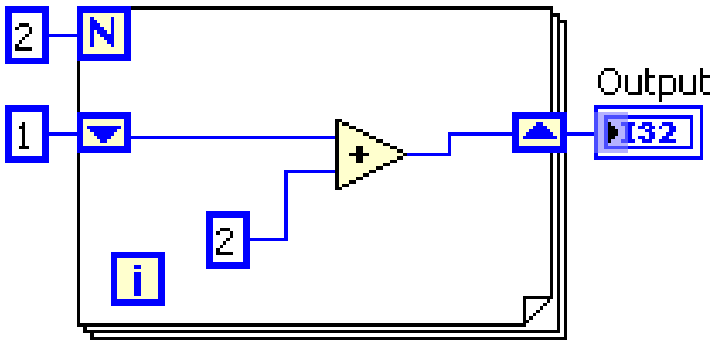
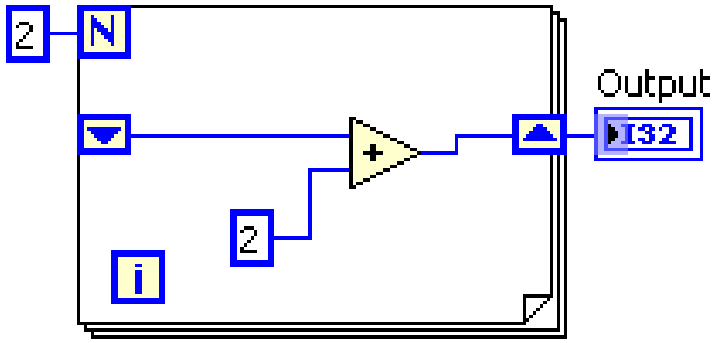
Shift Registers

- Right-click the border and select **Add Shift Register** from the shortcut menu.
- Right part of shift register stores data on completion of an iteration.
- Left part of shift register provides stored data at the beginning of the next iteration.



Initializing Shift Registers

Run once → VI finishes → Run again

Block Diagram	1st run	2nd run
Initialized Shift Register 	Output = 5	Output = 5
Not Initialized Shift Register 	Output = 4	Output = 8

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Default Value if Unwired

Default values vary by data type:

Data Type	Default Value
Numeric	0
Boolean	FALSE
String	Empty

Uninitialized shift registers use default values **for the first run**. At the beginning of next runs, they **remember** the last value they stored.

G. Case Structures

Parts of a Case Structure

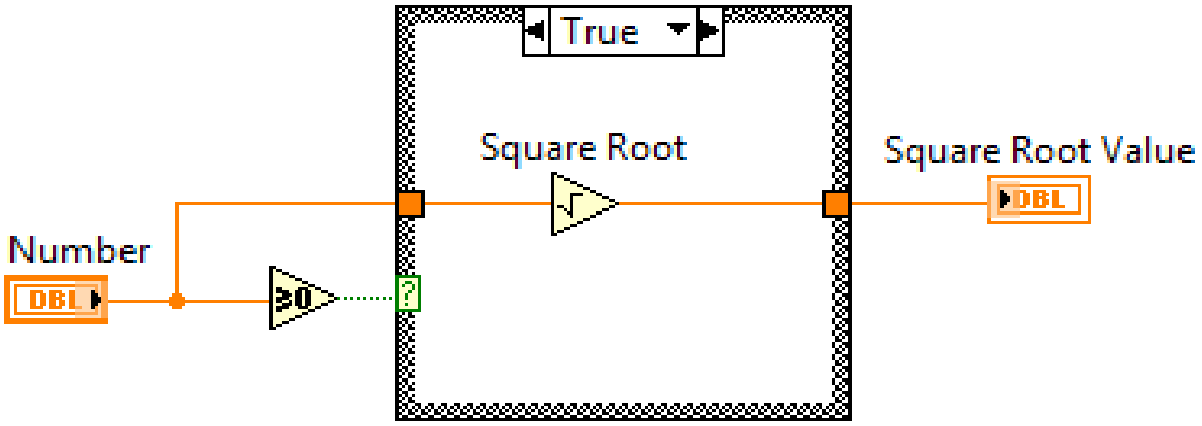
Enum Case Structures

Error Case Structures

Input and Output Tunnels

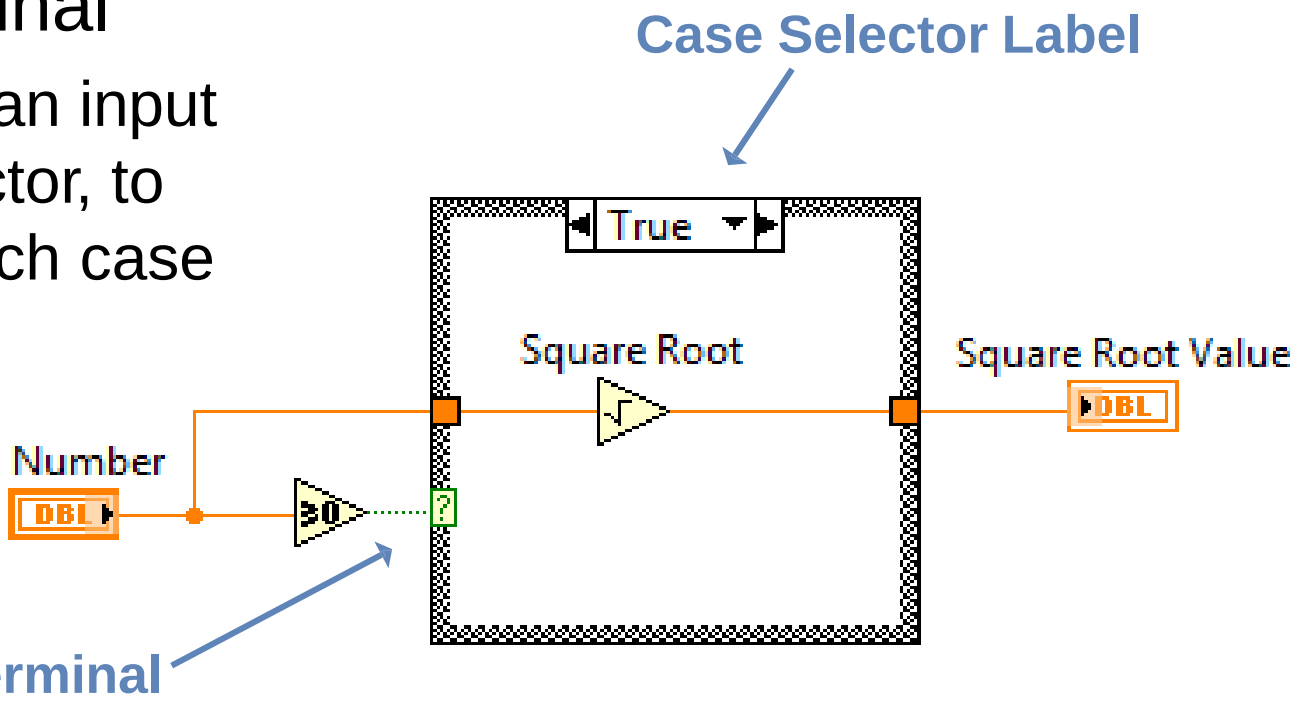
Case Structures

- Have two or more subdiagrams (cases).
- Use an input value to determine which case to execute.
- Execute and display only one case at a time.
- Are similar to **case** statements or **if...then...else** statements in text-based programming languages.



Case Structures

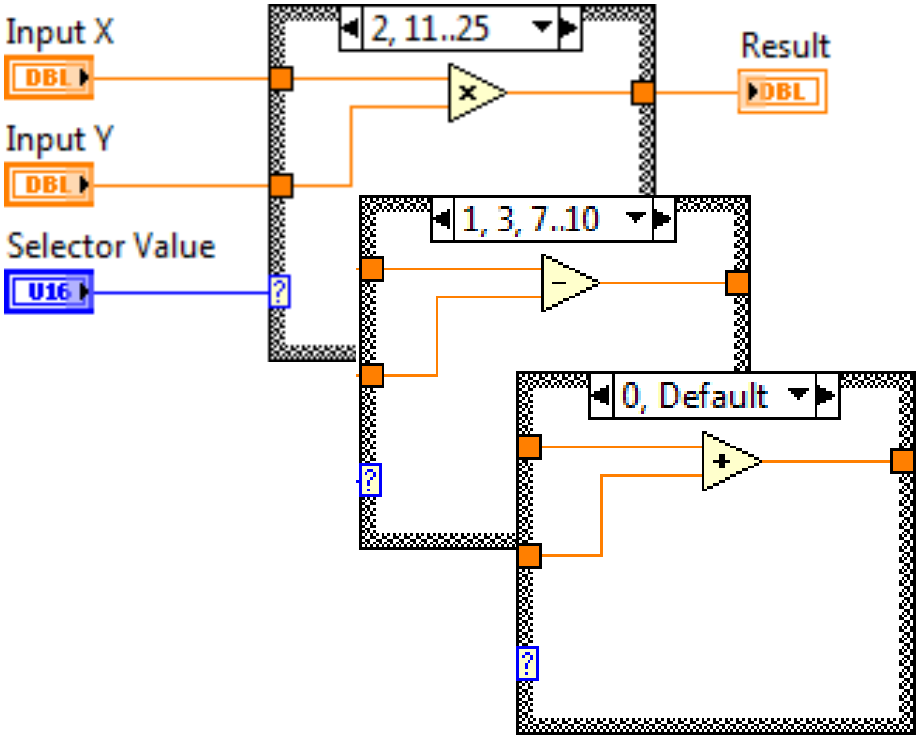
- Case Selector Label
  - Contains the name of the current case.
  - Has decrement and increment arrows.
- Selector Terminal
  - Lets you wire an input value, or selector, to determine which case executes.

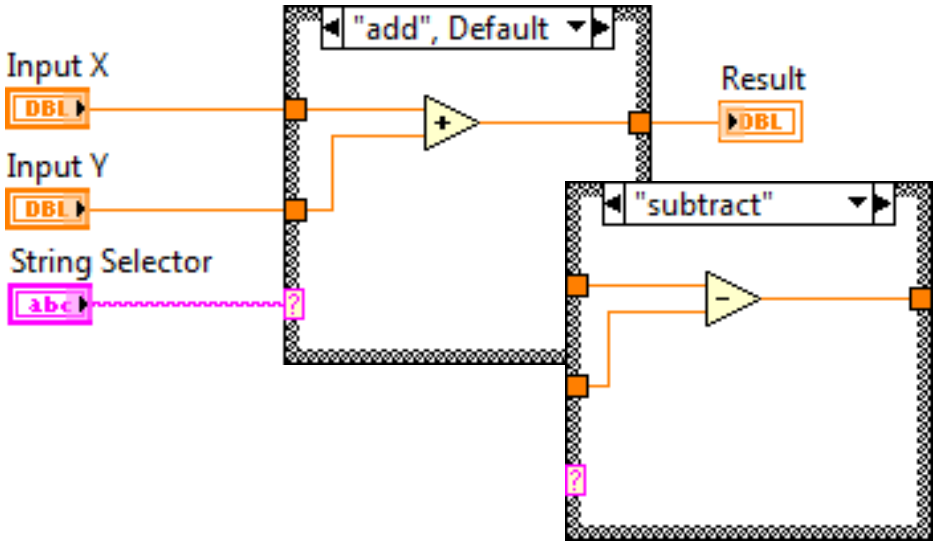


Case Structures

Selector terminal data types:

- Boolean
  - True case and False Case
- Error Cluster
  - Error Case and No Error Case
- Integer, string or enum
  - Structure can have any number of cases
  - Include a Default diagram to avoid listing every possible input value

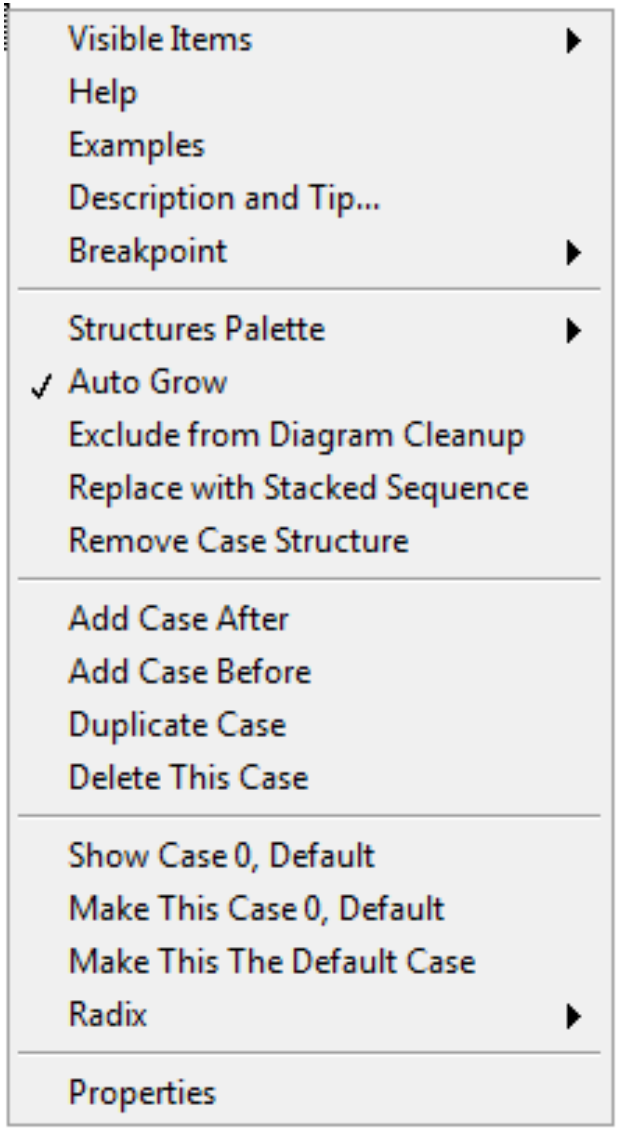




Shortcut Menu

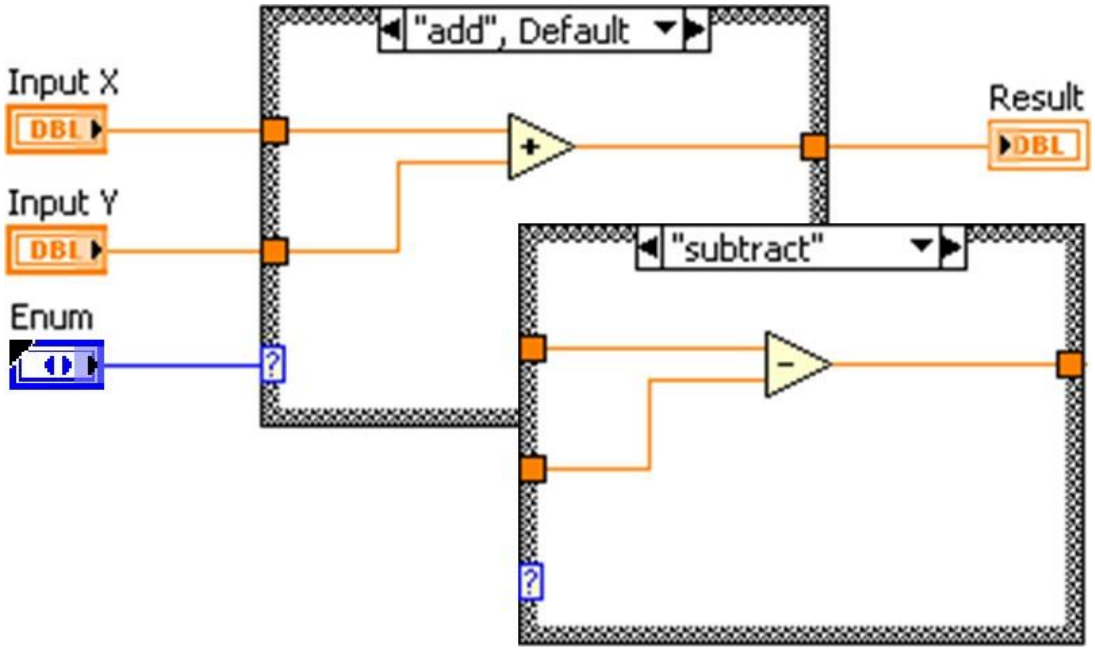
Use the shortcut menu of a Case structure to:

- Customize the structure and diagrams.
- Remove or replace the structure.
- Add, duplicate, remove or rearrange cases.
- Specify the Default case.
- Switch cases.



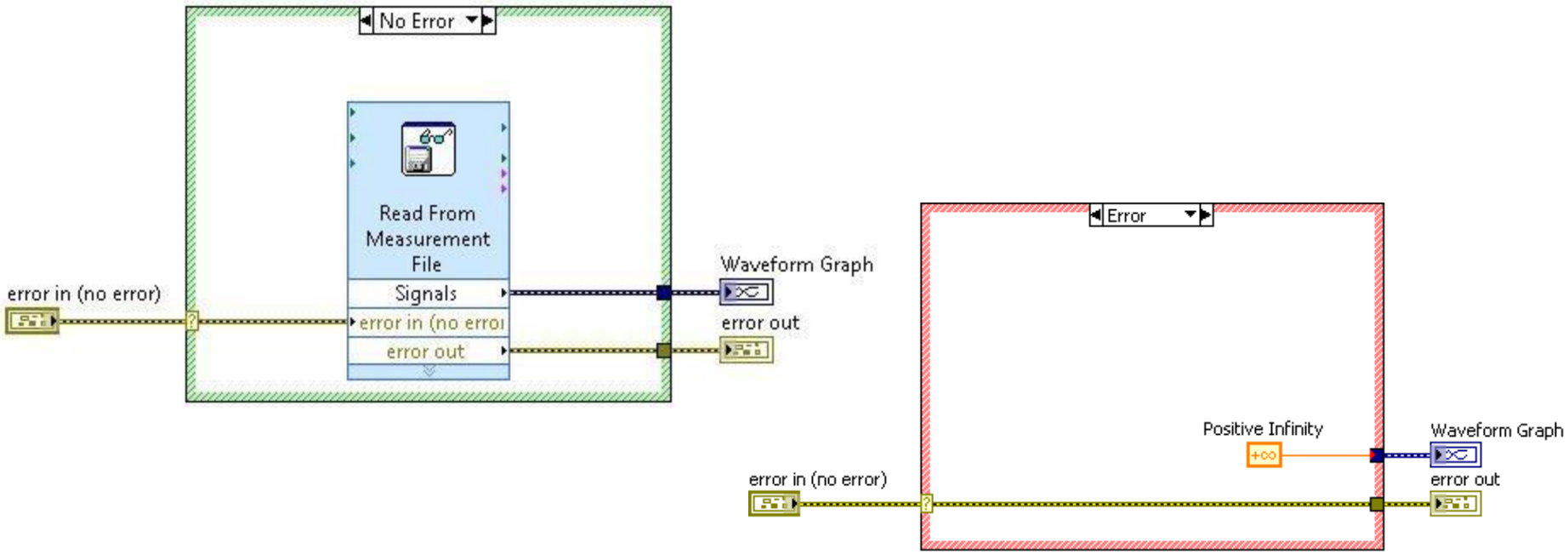
Enum Case Structure

- Gives users a list of items from which to select.
- The case selector displays a case for each item in the enumerated type control.



Error Case Structure

Use Case structures inside VIs to execute the code if there is no error and skip the code if there is an error.



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